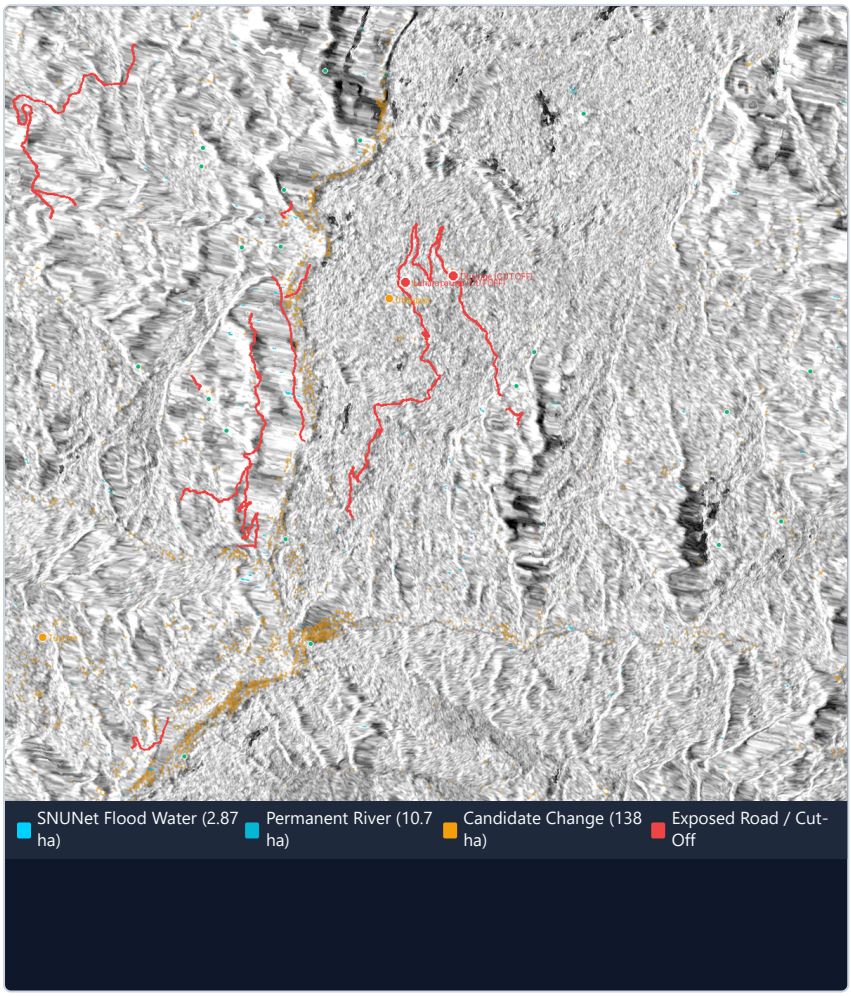
RAPID FLOOD DAMAGE & CONNECTIVITY SITUATION REPORT

Trishuli & Bhote Koshi River Corridor, Rasuwa / Nuwakot Districts, Nepal

Event Date: 2026-08-26 (Flash Flood / GLOF)
Analysis Run ID: flood_20261014_082338_45889d
Report Generated: 2026-10-14 17:30 UTC

SNUNET NEURAL INUNDATION	CANDIDATE RADAR CHANGE	EXPOSED HIGHWAY ASSETS	ISOLATED SETTLEMENTS
0.0287 km²	**1.3810 km²**	**12 Ways**	**2 Cut-Off**
2.87 ha (Deep Siamese Model)	138.1 ha (ΔVV < -3 dB Drop)	190.1 m Submerged (1,436 Fragments)	Laharepauwa, Dhunge (2 Detoured)



SATELLITE OBSERVATION BASELINE

Parameter	Analytical Specification
Sensor / Mode	Sentinel-1 RTC (C-band SAR, 10m grid)
Pre-Event Pass	**2026-08-24** 00:18 UTC (Orbit 19 Descending)
Post-Event Pass	**2026-09-05** 00:18 UTC (Orbit 19 Descending)
Analysis Extent	85.15°E–85.25°E, 27.95°N–28.05°N
Unpadded Valid AOI	**109.51 km²** (Padded grid: 118.05 km²)
Elevation Range	512 m to 3,485 m (Copernicus DEM 30m)

SETTLEMENT ACCESSIBILITY (CONSERVATIVE CORRIDOR CLOSURE)

Settlement	Baseline	Post-Event Status
Laharepauwa	5.31 km	**CUT-OFF (Corridor Closure)**
Dhunge	3.08 km	**CUT-OFF (Corridor Closure)**
Tupche	4.99 km	**DETOURED (+0.68 km)**
Uttargaya	7.37 km	**DETOURED (+4.42 km)**
23 Other Hamlets	Variable	**UNAFFECTED (incl. Betrawati)**

INFRASTRUCTURE EXPOSURE SUMMARY

Road Network: 12 unique OSM highway ways (totalling 190.1 m direct intersection) are inundated by the primary flood extent. Under the conservative corridor closure model (closing entire breached ways), 1,436 graph edge fragments (26.14 km) are removed. Strict fragment-level traversal severs 25 edges (383.9 m) with 0 cut-offs.
Bridge Infrastructure: 25 bridges assessed within the AOI; 0 confirmed bridge structural collapses under primary flood extent.

OPERATIONAL INTERPRETATION & SYNTHESIS

Highland Surge & Cross-Hazard Dynamics: On 26 August 2026, severe flash flooding and debris flows surged down the Trishuli corridor. SNUNet-ECAM deep Siamese segmentation identifies **2.87 hectares** of filtered standing flood water (41 polygons ≥ 500 m²; 15.76 ha raw raster) along river terraces. Meanwhile, classical radar drop (ΔVV < -3 dB, slope ≤ 15°) reveals **138.10 hectares** of physical surface alteration and debris scouring. In the **5.08 km²** shared evaluation area with Copernicus EMS activation **EMSR927** (AOI03 Bidur, an optical mass-movement grading product mapped on 27 Aug), classical radar drop captured **37.58 ha** of the debris track with **67.7% precision** and **21.4% IoU**. This comparison is strictly an independent cross-hazard spatial overlap analysis, not flood ground truth. Two settlements (Laharepauwa, Dhunge) lose road access under the conservative corridor-level closure scenario.

DATA QUALITY, UNCERTAINTIES & MANDATORY METHODOLOGICAL DISCLOSURES:
- **Corridor-Level Closure Scenario:** Cut-off hamlets reflect a conservative whole-way closure assumption (severing all 26.14 km of breached ways, including 25.95 km outside the 190.1 m flood intersection). Localized point traversal yields 0 cut-offs.
- **Building Inventory Completeness:** Pre-disaster OSM building footprints were query-capped at 1,000 features due to API limits. Zero exposed buildings reflects this mapped sample subset and does not guarantee structural safety of unmapped dwellings.
- **Observation & Sensor Semantics:** Sentinel-1 SAR was acquired on 5 Sept 2026 (9 days post-event). SNUNet targets standing water; EMSR927 maps optical mass-movement deposits.

Contains modified Copernicus Sentinel data 2026. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018. © OpenStreetMap contributors (ODbL). Neural model: Kuro Siwo Siamese U-Net++ (Bountos et al., NeurIPS 2024).